

CHAPTER -4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

INTRODUCTION

The Farm2Home database system is designed to connect farmers directly with customers through an online platform. It efficiently manages customers, farmers, products, orders, payments, deliveries, reviews, and administrators. Functional dependencies define the relationship between attributes in each table, while redundancy analysis and data anomaly detection help maintain data consistency, reduce duplication, and improve database performance.

1. CUSTOMER TABLE

Functional Dependency

Customer_ID → Name, Email, Phone, Address, Password

Redundancy Analysis

Customer information is stored only once in the Customer table. Orders reference only the Customer_ID, avoiding duplicate customer records.

Data Anomalies

Insert Anomaly: A new customer can register without placing an order.

Update Anomaly: Updating customer details is done only in the Customer table.

Delete Anomaly: Deleting an order does not remove customer information.

2. FARMER TABLE

Functional Dependency

Farmer_ID → Name, Email, Phone, Farm_Address, Password

Redundancy Analysis

Farmer details are stored only once. Products reference Farmer_ID instead of repeating farmer information.

Data Anomalies

Insert Anomaly: A farmer can register before adding products.

Update Anomaly: Updating farmer details is done only once.

Delete Anomaly: Deleting a product does not remove farmer information.

3. PRODUCT TABLE

Functional Dependency

Product_ID → Product_Name, Category, Price, Quantity, Description, Farmer_ID

Redundancy Analysis

Product details are stored only in the Product table. Orders reference Product_ID instead of storing duplicate product information.

Data Anomalies

Insert Anomaly: New products can be added before any customer places an order.

Update Anomaly: Product price or quantity is updated only once.

Delete Anomaly: Deleting an order does not remove product information.

4. ORDER TABLE

Functional Dependency

Order_ID → Customer_ID, Order_Date, Total_Amount, Order_Status

Redundancy Analysis

The Order table stores only order information. Customer details are referenced using Customer_ID.

Data Anomalies

Insert Anomaly: Orders can be created using an existing Customer_ID.

Update Anomaly: Order status can be updated without affecting other tables.

Delete Anomaly: Deleting an order does not remove customer information.

5. ORDER_ITEM TABLE

Functional Dependency

OrderItem_ID → Order_ID, Product_ID, Quantity, Price

Redundancy Analysis

This table links orders and products, preventing duplicate product information in the Order table.

Data Anomalies

Insert Anomaly: Products can be added to an order independently.

Update Anomaly: Quantity changes affect only the selected order item.

Delete Anomaly: Removing an order item does not delete the product.

6. PAYMENT TABLE

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Status, Amount

Redundancy Analysis

Payment information is stored separately from the Order table to avoid duplicate payment records.

Data Anomalies

Insert Anomaly: Payment details can be added after placing an order.

Update Anomaly: Payment status is updated only once.

Delete Anomaly: Deleting payment information does not remove the order.

7. DELIVERY TABLE

Functional Dependency

Delivery_ID → Order_ID, Delivery_Address, Delivery_Date, Delivery_Status

Redundancy Analysis

Delivery details are maintained in a separate table, preventing repeated delivery information.

Reason for No Data Anomalies

The Delivery table stores only delivery-related information and references the Order_ID using a foreign key. It does not duplicate customer or product data. Therefore, the table is already normalized, and significant insert, update, or delete anomalies do not occur.

8. REVIEW TABLE

Functional Dependency

Review_ID → Customer_ID, Product_ID, Rating, Review_Text, Review_Date

Redundancy Analysis

Review details are stored separately, allowing customers to review multiple products without duplicating customer or product information.

Reason for No Data Anomalies

The Review table stores only customer ratings and feedback. Each review is linked to a specific customer and product using foreign keys. Since it does not duplicate data from other tables, significant data anomalies are not present.

9. ADMIN TABLE

Functional Dependency

Admin_ID → Name, Email, Password

Redundancy Analysis

Administrator information is stored separately to avoid repeating login details across different tables.

Reason for No Data Anomalies

The Admin table contains only administrator login and profile information. It is an independent table with no repeated business data. Therefore, redundancy is minimal, and insert, update, or delete anomalies are not significant.

CONCLUSION

The Farm2Home database is designed using proper functional dependencies and normalized tables. Each entity stores only its relevant data, reducing redundancy and preventing insert, update, and delete anomalies. This design improves data consistency, maintains database integrity, and ensures efficient management of farmers, customers, products, orders, payments, deliveries, and reviews. It provides a reliable and scalable platform for the Farm2Home e-commerce application.